

**BANARAS HINDU UNIVERSITY**

B.A. (Hons.) — Semester IV Examination, 2023-24

**Paper No. MTB-401: Partial Differential Equations**

Time: 3 Hours

Full Marks: 70

Ref. No.: 076465

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

**Question 1.** (Compulsory — 2 marks each.)

- (a) Derive the PDE by eliminating arbitrary constants from the equation of a sphere.
- (b) Form a PDE by eliminating  $\phi$  from  $z = (x - y)\phi(x^2 + y^2)$ .
- (c) Solve:  $px + qy = az$ .
- (d) Find the complete integral of  $pq = p + q$ .
- (e) Find the particular integral of  $r + s - 2t = e^{x+y}$ .
- (f) Solve:  $(D - D')z = 0$ .
- (g) Solve:  $(2D - 3)(3D - 5D' - 7)^2z = 0$ .

**Question 2.**

- (a) Find the PDE by eliminating  $a, b, c$  from  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ . [7]
- (b) Find the complete integral, general integral, and singular integral of  $z = yp^2 - 4$ . [7]

**Question 3.**

- (a) Find the integral surface of  $x^2p + y^2q + z^2 = 0$  passing through the hyperbola  $xy = x + y$ ,  $z = 1$ . [7]
- (b) Solve:  $(D^2 - DD' - 2D'^2)z = (y - 1)e^x$ . [7]

**Question 4.**

- (a) Solve:  $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$ . [7]

- (b) Solve:  $ys + p = \cos(x + y) - y \sin(x + y)$ . [7]
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**Question 5.**

- (a) Solve:  $(x^2 D^2 + 2xy DD' + y^2 D'^2)z = x^m y^n$  where  $(m + n) \neq 0, 1$ . [7]
- (b) Solve:  $y^2(x - y)p + x^2(y - x)q = z^2(x^2 + y^2)$ . [7]
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**Question 6.**

- (a) Reduce  $r - t = 0$  to canonical form. [7]
- (b) Solve  $r = a^2 t$  by Monge's method. [7]
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**Question 7.**

- (a) Describe the two-dimensional heat equation by separation of variables. [7]
- (b) Solve the wave equation by separation of variables:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = x^2(25 - x^2)$$

[7]